

Osaid Khan

+1 (945) 304-5781 | khanosaid726@gmail.com | www.linkedin.com/in/-osaid-khan/ | github.com/ozzyozbourne

Dear Hiring Team,

It is with great pleasure that I submit my candidacy for the Software Development Engineer Graduate (Intent-Based Networking) - 2026 Start (BS/ MS) role with ByteDance. I bring over three years of professional experience building backend automation platforms, frontend workflow tooling, and production services, along with a strong academic foundation as a Master of Science in Computer Science graduate from Pace University, NY.

Through my experience at Sperse, I have built systems that align closely with ByteDance Networking's need for datacenter network management automation, lifecycle tooling, and operational stability. I built a Python, FastAPI, LangGraph, and LangChain control plane that routes 10,000+ users across 40+ specialized sub-agents with secure access control for high availability, scalability, and high-performance delivery. I also built platform products and tooling services in React Flow, TypeScript, HTML, and CSS for network planning, deployment, change, verification, and analytics capabilities, improving network delivery efficiency and stability with durable automated changes development. That combination of backend service design, intent-based workflow execution, and frontend operator tooling is the same engineering shape I would bring to ByteDance's work building platform products for the full lifecycle of network engineering.

I have also practiced the code and design discipline this role emphasizes. At Sperse, I built an agentic Code Review pipeline integrating OpenAI Codex with a custom Bitbucket MCP server, automating 25-30 PRs/week with strict code and design quality gates and reducing manual reviewer load by several hours. I created a FastMCP tooling service exposing 160+ backend actions to AI agents, turning intent-based requests into validated management and control protocols for monitored lifecycle automation, and built GitHub Actions CI/CD pipelines that use OpenTelemetry trace-based evals to flag latency, accuracy, and safety issues before production. These experiences map directly to ByteDance's focus on reliable software for automation, intelligence, verification, and stable delivery.

At Qualitest, I developed the backend and data-layer foundation needed for high-performance infrastructure software. I designed a Java and Spring Boot Storage Service with PostgreSQL and Redis that handled uploads, downloads, and deletions across S3, Google Cloud Storage, and MinIO while troubleshooting TCP/IP, DNS, and ARP network paths. I reduced query latency through SQL and NoSQL database programming and optimization in PostgreSQL and Redis, using window functions, materialized views, indexes, and React analytics, and orchestrated long-running backend jobs with AWS Step Functions and AWS Lambda to improve error recovery across 100K+ weekly executions. This work prepared me to reason across services, databases, protocols, and operational workflows when reliability matters.

I am drawn to ByteDance's mission to Inspire Creativity, Enrich Life, and to ByteStyle values such as Always Day 1, Seek Truth and Be Pragmatic, and Be Candid and Clear. The Networking team's work enables global products to operate with high availability and scale, which makes the mission concrete at the infrastructure layer. I would love to discuss how my experience with automation platforms, TypeScript workflow tooling, backend services, CI/CD, and database optimization can contribute to ByteDance Networking and its mission. I can be reached at +1 (945) 304-5781 or khanosaid726@gmail.com. Thank you for your time and consideration, I look forward to connecting!

Sincerely,
Osaid Khan